1

Introduction

Recommender systems (RSs) have undergone a substantial shift with the introduction of graph neural networks (GNNs) (Bronstein *et al.*, 2021; Hamilton, 2020; Scarselli *et al.*, 2009), which model users and items as nodes in a bipartite, undirected graph. Unlike classical collaborative filtering (CF) approaches, which learn representations solely from direct user-item interactions, graph-based models propagate information across multi-hop neighborhoods, capturing high-order connectivity patterns that can enrich the learned embeddings and improve accuracy. This shift has led to the development of graph CF methods, which redefine traditional CF paradigms by integrating graph representation learning techniques (Gao *et al.*, 2022; He *et al.*, 2020; Mao *et al.*, 2021; Wu *et al.*, 2023; Yu *et al.*, 2024). These foundational efforts established the viability of graph-based CF and motivated a large body of research seeking to refine GNN architectures for recommender systems.

Early works in this direction adapted general-purpose graph convolutional layers such as GraphSAGE (Hamilton *et al.*, 2017) and GCN (Kipf and Welling, 2017) to recommendation, often by placing message-passing layers on top of latent-factor models (Berg *et al.*, 2017; Wang *et al.*, 2019; Ying *et al.*, 2018). These foundational efforts established the viability of

graph-based CF and motivated a large body of research seeking to refine GNN architectures for recommender systems. Subsequent advancements explored ways to simplify message passing (Wu *et al.*, 2019), remove non-linearities (Chen *et al.*, 2020b; He *et al.*, 2020), and incorporate edge importance through reweighting schemes (Anelli *et al.*, 2022c; Mancino *et al.*, 2023) or attention mechanisms, as in GAT (Velickovic *et al.*, 2018), to prune noisy interactions and disentangle multiple user intents (Ma *et al.*, 2019; Wang *et al.*, 2021; Wang *et al.*, 2020; Zhang *et al.*, 2022). More recent lines of research have extended graph-based recommender systems in several directions. Semi-supervised approaches (Huang *et al.*, 2022) and contrastive learning frameworks (Khosla *et al.*, 2020) mitigate the sparsity of user–item graphs through carefully designed graph augmentations (Cai *et al.*, 2023; Lin *et al.*, 2022; Liu *et al.*, 2022; Wu *et al.*, 2021; Yu *et al.*, 2024; Yu *et al.*, 2022). Parallel efforts addressed fundamental limitations of message passing by tackling over-smoothing (Chen *et al.*, 2020a), over-squashing (Topping *et al.*, 2022), or both (Giraldo *et al.*, 2023), inspiring architectures that go beyond standard neighborhood aggregation (Mao *et al.*, 2021; Peng *et al.*, 2022b; Shen *et al.*, 2021; Wu *et al.*, 2024). More recently, some works have proposed to promote uniformity and alignment in graph CF (Yang *et al.*, 2023), while graph diffusion models (Vignac *et al.*, 2023) have also emerged as a promising direction for denoising interaction graphs and improving robustness in recommendation tasks (Jiang *et al.*, 2024b), and large language models (LLMs) have been exploited to enhance the node representations and edges within the historical user-item graph (Wei *et al.*, 2024b). To illustrate the evolution of this research area, Figure 1.1 presents a non-exhaustive timeline of representative GNN-based recommendation models.

Despite extensive research on GNN-based recommender systems, a fundamental question remains insufficiently answered: "Why do these models perform so well?" Existing work has explored both theoretical and experimental perspectives in an attempt to explain their success. Theoretically, several studies suggest that graph convolution operations act as smoothing filters on node features, enabling GNNs to capture latent structural patterns in recommendation graphs (Shen *et al.*, 2021). More recent research has highlighted issues such as over-smoothing and

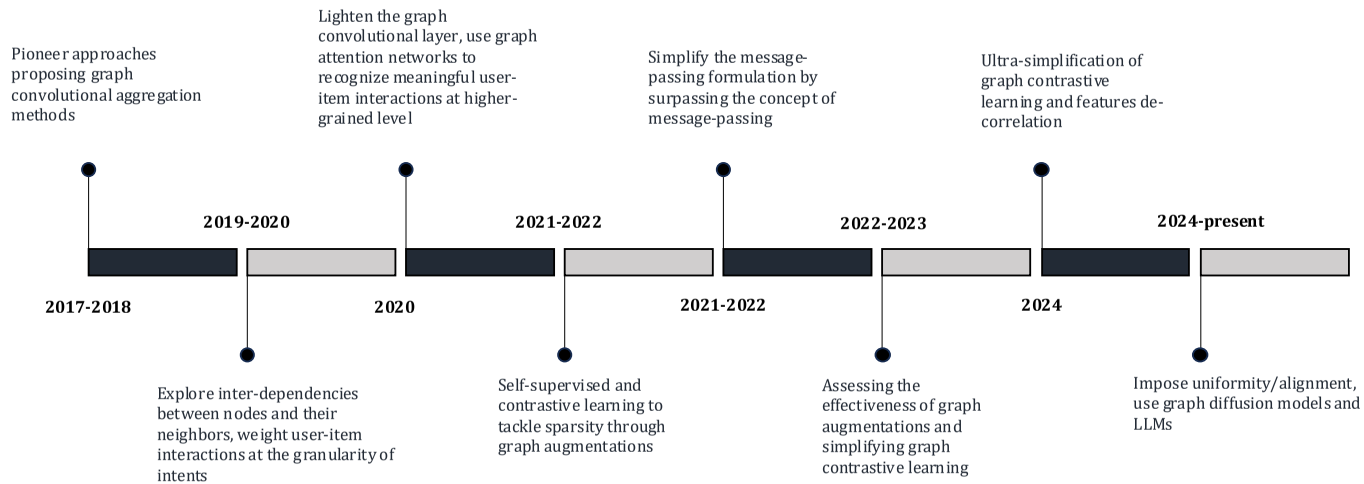


Figure 1.1: A non-exhaustive timeline of representative approaches in graph collaborative filtering (CF) by grouping solutions depending on the adopted graph representation learning strategies.

over-correlation, motivating adaptive techniques designed to mitigate these effects (Wu *et al.*, 2024). Experimentally, efforts have focused on modularizing GNN pipelines and generalizing architectures across recommendation scenarios (Wang *et al.*, 2022b); then, some works have compared both GNN-based (Anelli *et al.*, 2023b) and GNN session-based (Shehzad and Jannach, 2025) approaches with other carefully-tuned recommender systems, outlining performance pitfalls which are not commonly outlined in the original works; finally other authors identified node representation learning and neighborhood exploration as critical components, studying the performance of graph-based CF beyond accuracy metrics (Anelli *et al.*, 2023a; Anelli *et al.*, 2022b).

However, these investigations largely overlook a crucial factor: the role of graph topology in shaping GNN performance. Ultimately, answering the question "Why do GNN-based recommender systems work so well?" requires addressing a deeper one: "Which topological characteristics of user–item graphs do these models capture, and do those properties influence their performance?"

Our intuition is supported by recent developments in graph representation learning. A growing line of work has begun to interrogate how structural properties (such as topology and node features) affect GNN performance (Abu-El-Haija *et al.*, 2019; Castellana and Errica, 2023; Klicpera *et al.*, 2019; Luan *et al.*, 2019; Rong *et al.*, 2020; Wei *et al.*, 2022; Wu *et al.*, 2019; Xu *et al.*, 2018; Yang *et al.*, 2022; Zhao and Akoglu, 2020). For instance, it is now well established that GNNs excel in node classification tasks on graphs with high homophily (Zhu *et al.*, 2020). Likewise, recent studies show that node degree can significantly influence GNN performance in both node classification and link prediction tasks (Wang *et al.*, 2024c). Additionally, in recommender systems, existing studies have hinted that dataset characteristics (such as sparsity) can affect performance (Adomavicius and Zhang, 2012; Deldjoo *et al.*, 2020; Malitesta *et al.*, 2024); more recently, the works by Gioia *et al.* (2026a) and Gioia *et al.* (2026b) have investigated how spectral graph models can differently shape the topology of the user-item graph, while the study conducted by Fioretti *et al.* (2025) has assessed the impact of LLM augmentations on the user-item graph from a topological perspective. Yet, a systematic investigation of topological

factors specifically in recommendation is still missing, leaving a key aspect of GNN-based CF unexplored.

In light of the above, our work offers a topology-centered perspective on GNN-based recommendation, systematically relating the performance of representative models to the structural properties of the user–item graphs on which they operate. Building on this analysis, we derive a set of take-home messages on when and why specific strategies are effective, and we identify outstanding challenges and future research directions for topology-aware recommender systems. In the following, we outline the structure of the monograph (Section 1.1), the main notation and terminology (Section 1.2), and selected datasets and GNN models to exemplify our analysis (Sections 1.3 and 1.4).

1.1 Structure of this monograph

This work begins by introducing basic notions about graph neural networks (Section 2) to provide the **foundations** to understand and categorize graph-based recommendation algorithms (Section 3). This allows us to present a clear **taxonomy** of the methods and a **pipeline** of the recommendation process. This includes an overview of how GNN-based recommender systems have evolved, from early approaches that adapted shallow graph representation learning techniques to more recent trends involving semi-supervised and contrastive learning, to approaches translating graph-based recommendation to the spectral domain to address problems such as over-smoothing.

Sections 4 and 5 define several **dataset characteristics**, either classical or topological. How data is structured, shaped, or distributed across various dimensions impacts differently recommendation algorithms depending on their nature, as observed in the literature. Hence, we here divide the presentation of these dataset characteristics according to their topological interpretation.

Then, we examine how various **(topological) dataset characteristics** impact the recommendation performance of GNN algorithms (Section 6). This is achieved by re-interpreting popular topological graph properties under the lens of recommendation and highlighting their (explicit) presence within the models' formulations. Afterwards,

in Section 7, we present an **explanatory framework** that involves generating a large set of small graph recommendation datasets encompassing a wide range of topological structures and testing on them the recommendation performance of selected GNN-based models. Hence, **performance explanations** are derived by jointly considering the models' results and the dataset properties. Building upon the topological re-interpretations provided on classical GNNs-based recommendation, in Section 8, we offer an overview of recent trends in graph-based recommendation, more specifically those involving **spectral** methodologies and **LLMs**-based approaches, underlining topological implications of such novel strategies.

Based on these conclusions, Section 9 presents the most important **take-home messages** to be considered when dealing with GNN-based recommendation algorithms. This, together with the **challenges** and **future directions** included in Section 10, conclude the monograph and paves the way for further research, emphasizing issues and problems with more potential to be considered interesting for the community and industry practitioners.

1.2 Notations and terminology

We introduce the useful notation and terminology for this monograph. While this section provides most of the notations adopted in the following pages, we highlight that each section in this monograph might sometimes contain more specific notations tailored to a particular formulation that is not covered here.

We use boldface capital letters to represent multi-dimensional vectors (e.g., $\mathbf{A}$), lowercase letters for constants and coefficients (e.g., t), and Greek uppercase/lowercase letters for classical and topological dataset properties, respectively (we summarize them in Table 1.1). Note that, for the degree distribution only, we do not introduce a symbolic notation, as its behavior is more effectively analyzed through its empirical shape. For functions, we indicate them in a textual, human-readable format, such as LeakyReLU$(\cdot)$ or $\exp(\cdot)$ for the LeakyReLU activation function (Xu *et al.*, 2015) and the exponential function, respectively. Moreover, we use the calligraphic notation for loss functions and sets, for instance $\mathcal{L}_{\text{BPR}}$

and $\mathcal{U}, \mathcal{I}$ to refer to the Bayesian-personalized ranking loss function (BPR, introduced by Rendle *et al.* (2009)) and the sets of users and items, respectively.

Regarding vectors, we indicate specific entries of these vectors with subscripts, such as A_{ij} to indicate the i-th row and j-th column (the boldface notation is removed here as we refer to a single scalar element); we adopt the "*" notation to refer to all entries for one dimension (e.g., $\mathbf{A}_{i*}$ refers to the i-th row and all columns for matrix $\mathbf{A}$). Conversely, we use superscripts mainly to refer to updated versions of vectors during the message passing algorithm, such as $\mathbf{E}_u^{(l)}$ to refer to the updated version of the vector $\mathbf{E}_u$ after l message passing layers. Additionally, we adopt the hat and tilde notations (e.g., $\hat{\mathbf{E}}_u$ and $\tilde{\mathbf{A}}$) to indicate vectors that have been completely updated to compute a prediction, and a normalized vector (as in the case of the normalized adjacency matrix), respectively. Finally, we leverage the operator $[\cdot]$ to refer to a specific view of a vector, as in the case of graph contrastive learning (e.g., $\mathbf{E}_u[t]$ to indicate the t-th view of vector $\mathbf{E}_u$).

Terminology-wise, we highlight that we refer to the whole family of models covered in this monograph in different variants, such as "graph-based recommender systems", "GNNs-based recommender systems", or "graph CF recommender systems".

1.3 Datasets referenced in this monograph

As one of the main rationales behind this monograph is to offer a dataset-centric perspective on graph-based recommendation (with specific focus on topology), the next sections introduce both classical and topological properties of the recommendation dataset. To complement the theoretical description of such characteristics, we select popular datasets from the recommendation literature, and compute values for these properties, thus providing the reader with also a practical illustration. For the selection, we follow the dataset list collected in (Mancino *et al.*, 2025) and limit ourselves to the six most popular datasets in the recommendation literature.

Table 1.1: The table summarizes the data characteristics analyzed in this work, the symbols used to refer to them, and their shorthand notation.

Type	Characteristics	Symbol	Shorthand
Classical	Space size	Ψ	SpaceSize
	Shape	Φ	Shape
	Density	Δ	Density
	Sparsity	Λ	Sparsity
	Gini user	$\mathrm{K}_{\mathcal{U}}$	Gini-U
	Gini item	$\mathrm{K}_{\mathcal{I}}$	Gini-I
Topological	Average degree user	$\sigma_{\mathcal{U}}$	AvgDegree-U
	Average degree item	$\sigma_{\mathcal{I}}$	AvgDegree-I
	Avg. clustering coefficient user	$\gamma_{\mathcal{U}}$	AvgClustC-U
	Avg. clustering coefficient item	$\gamma_{\mathcal{I}}$	AvgClustC-I
	Degree assortativity user	$\rho_{\mathcal{U}}$	Assort-U
	Degree assortativity item	$\rho_{\mathcal{I}}$	Assort-I
	Degree distribution	–	DegreeDistr

- **MovieLens 1M**[1] (Harper and Konstan, 2016). A widely used benchmark dataset containing movie ratings collected from the MovieLens movie recommendation platform.
- **Yelp**[2]. The Yelp Open Dataset contains interactions between users and businesses, where user feedback is provided in the form of written reviews.
- **Amazon Beauty**[3] (Hou *et al.*, 2024). A large-scale dataset collected in 2023 from the Amazon website, containing user reviews for products in the Beauty category.
- **Amazon Books**[4] (Hou *et al.*, 2024). A large-scale dataset collected in 2023 from the Amazon website, containing user reviews for products in the Books category.

[1] https://grouplens.org/datasets/movielens/
[2] https://business.yelp.com/data/resources/open-dataset/
[3] https://amazon-reviews-2023.github.io/
[4] https://amazon-reviews-2023.github.io/

Table 1.2: Statistics of the recommendation datasets referenced in this monograph after filtering, ordered by their total number of interactions (the number of edges in the corresponding bipartite graph).

Dataset	# Users	# Items	# Interactions
MovieLens	5,611	2,932	983,705
Yelp	14,691	12,482	819,986
Amazon Beauty	8,425	11,509	304,396
Amazon Books	3,250	2,450	141,331
Gowalla	2,426	2,195	131,715
LastFM	1,601	951	53,491

- **Gowalla**[5] (Cho *et al.*, 2011). A dataset derived from the location-based social network Gowalla, containing user check-ins recorded between Feb. 2009 and Oct. 2010.
- **Last.fm**[6] (Cantador *et al.*, 2011). A dataset derived from the Last.fm music platform, containing users' listening preferences and interactions with musical artists.

To simplify the analysis and reduce noise, we filtered the datasets by applying an iterative user–item k-core with different depths: 15 for Amazon Beauty and LastFM, 25 for Amazon Books, MovieLens, Gowalla, and Yelp. The statistics obtained after filtering are reported in Table 1.2, where the datasets are ordered by their total number of interactions, from the highest to the lowest.

1.4 Models referenced in this monograph

Alongside the selected dataset characteristics, we also consider a list of eleven graph-based recommendation approaches from the literature, to re-interpret their formulations under the topological lens. The motivation behind this selection is twofold. On the one side, these models constitute a comprehensive (but non-exhaustive) group of approaches

[5] https://snap.stanford.edu/data/loc-gowalla.html
[6] https://grouplens.org/datasets/hetrec-2011/

spanning well-established (i.e., counting several citations on Google Scholar) and more recent solutions, encompassing diversified strategies from shallow GNNs architectures to more refined techniques involving graph contrastive learning. On the other side, prior research (such as the work by Anelli *et al.* (2023b)) has evidenced that most of these approaches are constantly tested as strong graph-based baselines for novel proposed methodologies. In the following, we briefly summarize their architecture and rationales, displaying their publication year and venue in Table 1.3, where we sort models according to the chronological order of their publication.

- **NGCF.** Neural graph collaborative filtering (Wang *et al.*, 2019) is one of the pioneer approaches in graph-based recommendation. The model implements a GCN-like message passing, where each propagation layer takes also into account the inter-dependencies between the ego and neighborhood nodes.

- **DGCF.** Disengangled graph collaborative filtering (Wang *et al.*, 2020) proposes to divide user and item vectors into intents to represent the reasons behind each recorded user-item interaction and weight their importance.

- **LightGCN.** Light graph convolutional network (He *et al.*, 2020) simplifies the GCN message passing from previous approaches by removing the features transformation and non-linearities from the message passing procedure, while still reaching improved recommendation performance.

- **SGL.** Self-supervised graph learning (Wu *et al.*, 2021) builds different augmented views of the user-item graph and applies a contrastive procedure between them to limit the sparsity problem of the recommendation dataset.

- **UltraGCN.** Ultra simplification of graph convolutional network (Mao *et al.*, 2021) tailors crucial issues in graph learning, such as over-smoothing and usually-unexplored node relationships (such as item-item). To this end, it proposes a mathematical simplification of the GCN message passing layer that (in one step)

approximates an ideal infinite-layer message passing, and specific loss components to further address over-smoothing and consider item-item interactions.

- **GFCF.** Graph filter-based collaborative filtering (Shen *et al.*, 2021) re-interprets graph convolution in recommendation under the lens of graph signal processing, by formulating a unified graph filtering framework for most approaches in CF. On such a basis, it proposes a closed-form solution that performs better than other trainable approaches in the literature.

- **SVD-GCN.** The authors in (Peng *et al.*, 2022b) applies singular value decomposition (SVD) to the user-item interaction matrix to calculate, through a closed-form solution, vector representations for users and items. On such a basis, they design additional loss function components (besides BPR) that take into account user-user and item-item similarities.

- **SimGCL.** Simplification of graph contrastive learning (Yu *et al.*, 2022) questions the role of traditional graph augmentations (such as node and edge dropout), proposing a new augmentation strategy applied on node vectors during the message passing procedure. The approach further implements a uniformity loss on the learned representations.

- **LightGCL.** The authors in (Cai *et al.*, 2023) design a double message passing procedure in graph contrastive learning, one acting locally on the perturbed graph structure, and the other acting globally by truncating the SVD representation of users and items. The two views are contrasted in the loss function, which also implements a hinge loss unlike many other solutions leveraging BPR.

- **GraphAU.** The model proposed in (Yang *et al.*, 2023) re-adapts the principles of uniformity and alignment to graph-based recommendation by setting up a trade-off loss between uniformity and alignment on the user-item graph structure.

- **XSimGCL.** Extremely simple graph contrastive learning (Yu *et al.*, 2024) surpasses the traditional concept of graph contrastive learning in recommendation by leveraging the same augmented node view for the final recommendation task. Moreover, the contrastive loss accounts for two versions of the same node vectors computed at different propagation layers in the message passing.

Table 1.3: GNN-based recommendation models referenced in this monograph, sorted by publication year.

Model	Paper	Venue
NGCF	Neural Graph Collaborative Filtering (Wang *et al.*, 2019)	SIGIR'19
DGCF	Disentangled Graph Collaborative Filtering (Wang *et al.*, 2020)	SIGIR'20
LightGCN	LightGCN: Simplifying and Powering Graph Convolution Network for Recommendation (He *et al.*, 2020)	SIGIR'20
SGL	Self-supervised Graph Learning for Recommendation (Wu *et al.*, 2021)	SIGIR'21
UltraGCN	UltraGCN: Ultra Simplification of Graph Convolutional Networks for Recommendation (Mao *et al.*, 2021)	CIKM'21
GFCF	How Powerful is Graph Convolution for Recommendation? (Shen *et al.*, 2021)	CIKM'21
SVD-GCN	SVD-GCN: A Simplified Graph Convolution Paradigm for Recommendation (Peng *et al.*, 2022b)	CIKM'22
SimGCL	Are Graph Augmentations Necessary? Simple Graph Contrastive Learning for Recommendation (Yu *et al.*, 2022)	SIGIR'22
LightGCL	LightGCL: Simple Yet Effective Graph Contrastive Learning for Recommendation (Cai *et al.*, 2023)	ICLR'23
GraphAU	Graph-based Alignment and Uniformity for Recommendation (Yang *et al.*, 2023)	CIKM'23
XSimGCL	XSimGCL: Towards Extremely Simple Graph Contrastive Learning for Recommendation (Yu *et al.*, 2024)	TKDE'24